

Lecture 6: Atmospheres II

Clouds, Weather, & Climate

Planetary Systems

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Learning objectives

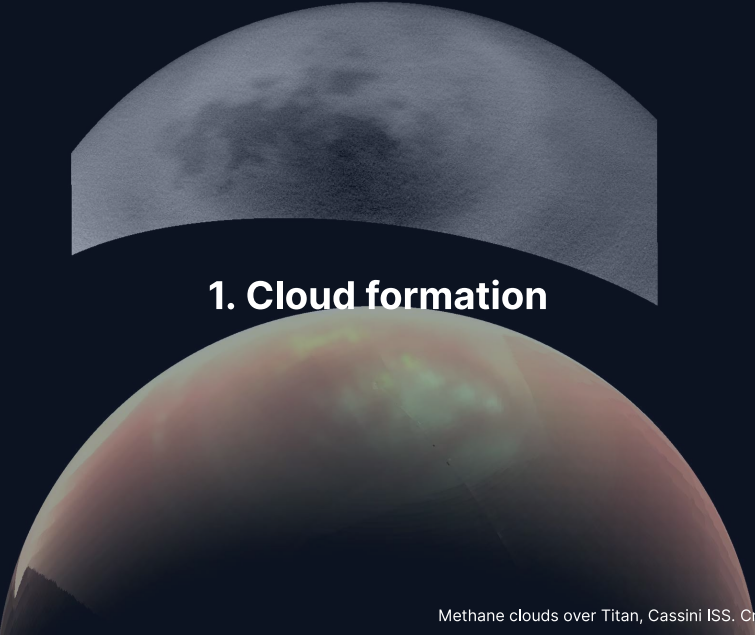
By the end of this lecture, you will be able to:

- Derive the Clausius-Clapeyron equation and predict cloud formation
- Explain circulation using the Coriolis effect and geostrophic balance
- Describe weather phenomena throughout the solar system
- Discuss climate evolution and the faint young Sun paradox
- Explain the carbonate-silicate cycle as a climate thermostat

Recap: Lecture 5

- Hydrostatic equilibrium $dP = -\rho g dz$ and scale height $H = k_B T / (\mu m_u g)$ set atmospheric structure
- Radiative equilibrium and greenhouse warming set surface temperatures ($T_s > T_{eq}$)
- Convection drives the adiabatic lapse rate $\Gamma_d = g/c_p$ in the lower atmosphere

Today: What happens when vapour condenses? How do winds blow? Why has Earth's climate been stable for 4 Gyr?



1. Cloud formation

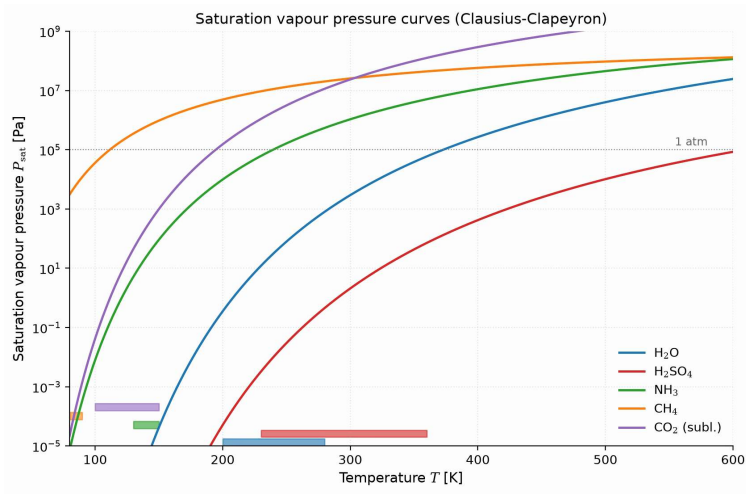
Methane clouds over Titan, Cassini ISS. Credit: NASA/JPL-Caltech/SSI

Saturation and condensation

- Every vapour has a **saturation vapour pressure** $P_{\text{sat}}(T)$
- When actual vapour pressure exceeds P_{sat} : air is **supersaturated**
- $P_{\text{sat}}(T)$ increases roughly exponentially with temperature
- **Relative humidity:**

$$\text{RH} = \frac{P_{\text{vapour}}}{P_{\text{sat}}(T)} \times 100\%$$

- RH = 100%: saturated; RH > 100%: condensation can occur
- Rising air cools $\rightarrow P_{\text{sat}}$ drops $\rightarrow$ reaches saturation $\rightarrow$ clouds



Saturation vapour pressure for the five main condensables. A single equation, evaluated per species, predicts every cloud deck in the solar system. Course figure.

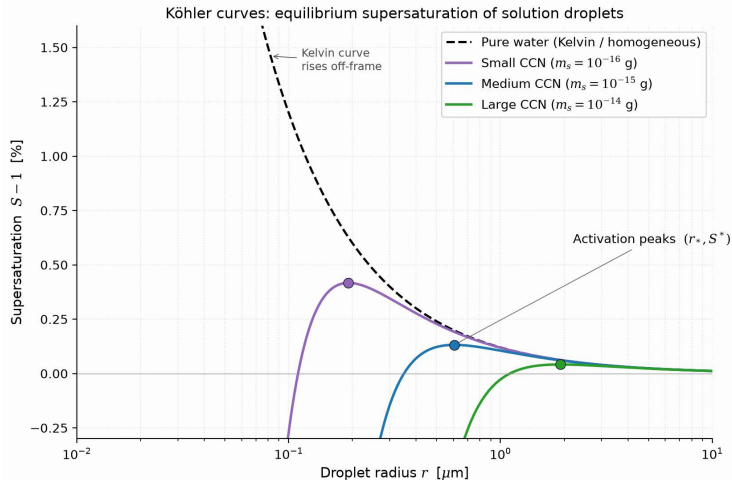
Nucleation: how droplets form

Even in supersaturated air, condensation needs a **seed**.

Homogeneous nucleation: Droplets from vapour alone; requires extreme supersaturation ($RH \gg 100\%$), very rare

Heterogeneous nucleation: Condensation onto pre-existing nuclei (dust, sea salt, soot); activates at only ~0.04–0.4% supersaturation

- Earth: oceanic/biogenic nuclei; Mars: dust; Giant planets: photochemical hazes



Köhler curves: equilibrium supersaturation of solution droplets. Real clouds nucleate heterogeneously, with aerosols lowering the barrier by two orders of magnitude. Course figure.

The lifting condensation level (LCL)

Rising air cools dry adiabatically ($\Gamma_d = g/c_p$) until $P_{\text{vapour}} = P_{\text{sat}}(T)$, i.e. until it reaches its **dew point** T_d .

The **lifting condensation level** (LCL) marks the **cloud base**.

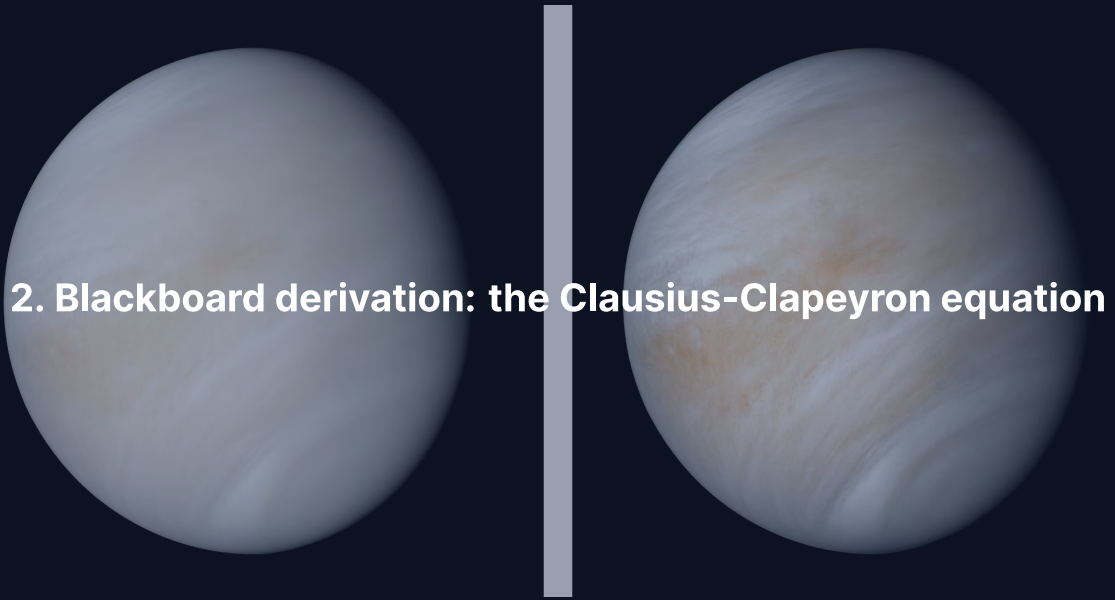
- Above the LCL, latent heat release reduces cooling to the moist adiabatic lapse rate ($\sim 5 \text{ K km}^{-1}$ vs. $\sim 9.8 \text{ K km}^{-1}$ dry)
- Latent heat fuels buoyancy, powering deep convective storms up to the tropopause

Cloud types of the solar system

What condenses depends on temperature and composition:

Body	Cloud species	Cloud base altitude
Earth	H ₂ O	~1–2 km
Venus	H ₂ SO ₄	~48 km
Mars	CO ₂ ice, H ₂ O ice	10–100 km
Titan	CH ₄ , C ₂ H ₆	~8–30 km
Jupiter	NH ₃ , NH ₄ SH, H ₂ O	layered, 0.5–7 bar

The **Clausius-Clapeyron equation** governs every cloud deck.



2. Blackboard derivation: the Clausius-Clapeyron equation

Deriving the Clausius-Clapeyron equation

Goal: Derive the exponential dependence of saturation vapour pressure on temperature from thermodynamic phase equilibrium.

Setup: phase equilibrium.

Along the coexistence curve, liquid and vapour are in equilibrium:

$$g_\ell(T, P) = g_v(T, P)$$

where g is the specific Gibbs free energy.

Moving along the coexistence curve by (dT, dP) :

$$dg_\ell = dg_v$$

To the board.

The Clausius-Clapeyron equation

$$P_{\text{sat}}(T) = P_{\text{ref}} \exp \left[-\frac{L_v}{R_v} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right]$$

Credit: Derivation follows Catling (2017), *Atmospheric Evolution on Inhabited and Lifeless Worlds*, Ch. 1

- P_{sat} depends **exponentially** on temperature
- The sensitivity is controlled by L_v/R_v (units of K)
- Large $L_v/R_v \rightarrow$ very sensitive to temperature changes
- This equation applies to **any** vapour $\leftrightarrow$ condensed phase transition

Worked example: water on Earth

For water vapour:

L_v	$2.50 \times 10^6 \text{ J kg}^{-1}$
M	$0.018 \text{ kg mol}^{-1}$
$R_v = R^* / M$	$462 \text{ J kg}^{-1} \text{ K}^{-1}$
T_{ref} (triple point)	273 K
P_{ref} (triple point)	611 Pa
L_v / R_v	$\approx 5400 \text{ K}$

At $T = 293 \text{ K}$ (20°C):

$$P_{\text{sat}} = 611 \exp\left[-5400\left(\frac{1}{293} - \frac{1}{273}\right)\right] = 611 \times 3.86 \approx 2360 \text{ Pa}$$

Measured value: 2300 Pa; excellent agreement.

Condensing species in the solar system

Species	L_v (kJ kg ⁻¹)	R_v (J kg ⁻¹ K ⁻¹)	L_v/R_v (K)	T_{cond} (K)	Where
H ₂ O	2500	462	5400	200–280	Earth, Mars
H ₂ SO ₄	540	85	6400	230–360	Venus
NH ₃	1370	488	2800	130–150	Jupiter, Saturn
CH ₄	510	519	980	80–90	Titan, Uranus, Neptune
CO ₂	571	189	3020	100–150	Mars

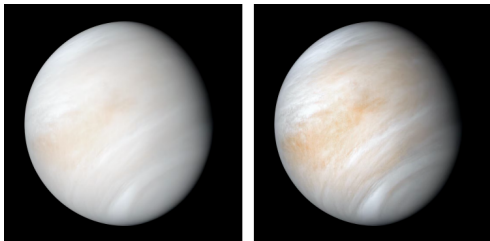
Steep L_v/R_v curves sharpen cloud bases (Venus); flat curves broaden decks (Titan).



3. Clouds of the solar system

Water-ice clouds over Mars, Mars Odyssey THEMIS (2025). Credit: NASA/JPL-Caltech/ASU

Venus: sulfuric acid clouds

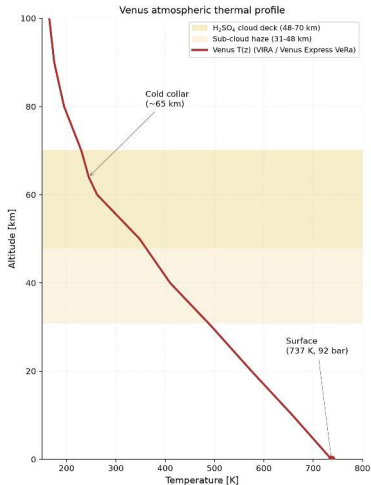


Credit: NASA/JPL-Caltech (1974)

- Concentrated H_2SO_4 clouds span 48–70 km altitude
- Photochemistry from outgassed SO_2 and photolytic O:

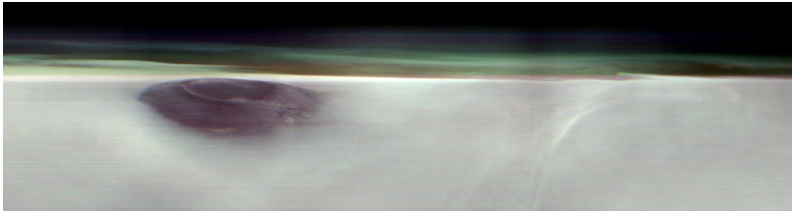


- Unidentified UV absorber creates upper-cloud tracking features
- Sub-cloud haze extends down to ~31 km



Venus thermal structure, VIRA / Venus Express VeRa. Concentrated H₂SO₄ clouds form a 22 km deck at 48–70 km altitude where Clausius-Clapeyron allows them. Course figure.

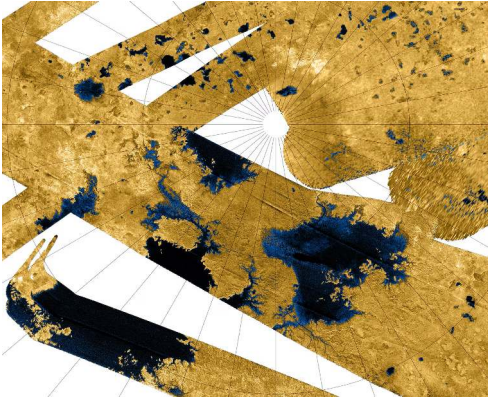
Mars: dust and ice clouds



Water-ice clouds over Tharsis limb. Credit: NASA/JPL-Caltech/ASU (Mars Odyssey THEMIS, 2 May 2025)

- CO_2 **ice**: mesospheric, 50–100 km, where T falls below the local CO_2 frost point (148 K at the 6 mbar surface, lower aloft)
- H_2O **ice**: 10–30 km over Tharsis volcanoes
- Mineral dust acts as condensation nuclei
- Radiative heating drives positive dust feedback

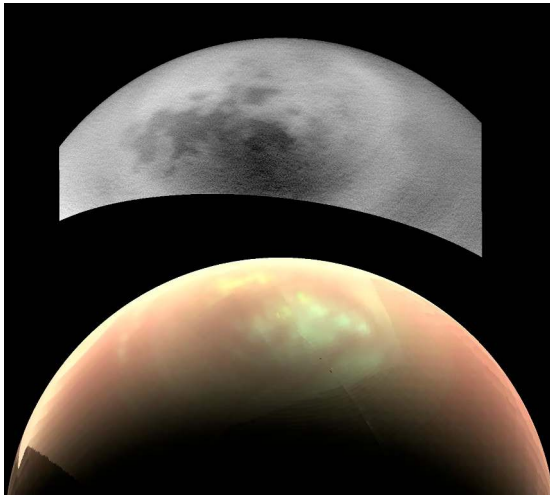
Titan's methane cycle: an Earthlike analogue



Credit: NASA/JPL-Caltech/ASI/USGS (Cassini)

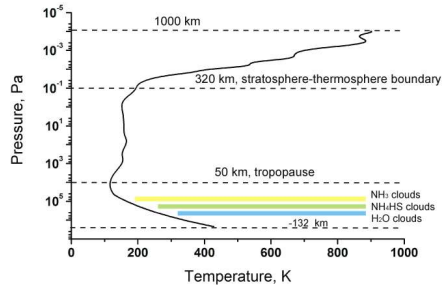
Active hydrological cycle with CH_4 near its triple point:

- Surface at 94 K and 1.5 bar supports liquid methane seas
- **Kraken Mare:** $\sim 400,000 \text{ km}^2$, locally $>100 \text{ m}$ deep
- Methane rain carves river channels and fills polar lakes
- Driven by slow 30-year seasonal solar cycling



Methane clouds at Titan's mid-southern latitudes, Cassini ISS. Methane rain at 94 K drives the only active hydrological cycle beyond Earth with polar convective storms. Credit: NASA/JPL-Caltech/SSI.

Giant planets: layered cloud structure



Credit: Wikimedia, CC BY-SA 3.0

Vertically layered, predicted by Clausius-Clapeyron:

1. NH_3 **ice** (top): 130–150 K, 0.5–1 bar
2. NH_4SH : 200–240 K, 2–3 bar ($\text{NH}_3 + \text{H}_2\text{S}$)
3. H_2O **ice/liquid** (deep): 270–300 K, 5–7 bar

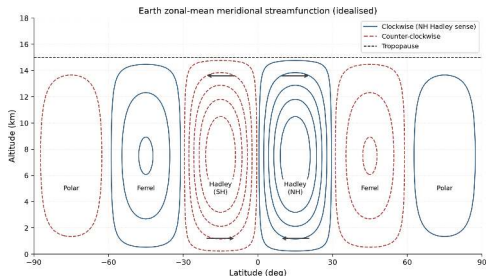
Latent heat release from deep water clouds fuels moist convection.

A satellite view of Earth from space, showing the curvature of the planet and a prominent jet stream cirrus cloud pattern. The clouds are white and wispy, stretching across the blue ocean and dark landmasses. The text "4. Atmospheric dynamics" is overlaid in the center.

4. Atmospheric dynamics

Jet-stream cirrus from orbit. Credit: NASA/JSC

The Hadley cell



Credit: Course figure

1. Solar heating at the equator drives buoyant rising air
2. Air flows poleward aloft, cooling radiatively
3. Sinking air at $\sim 30^\circ$ latitude creates subtropical desert belts
4. Equatorward surface return flow forms the trade winds

The primary tropical heat transport loop from equator to mid-latitudes.

The Coriolis effect

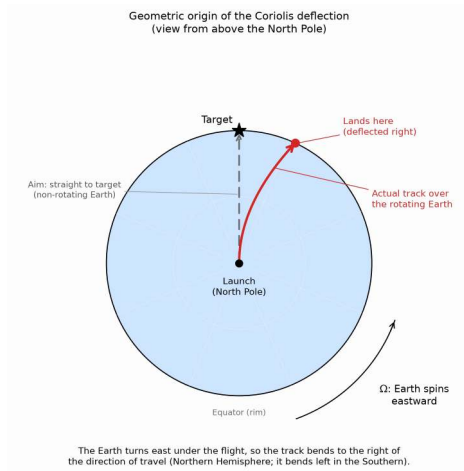
On a rotating planet, moving air is deflected right in the Northern Hemisphere, left in the Southern.

The **Coriolis parameter**:

$$f = 2\Omega \sin \phi$$

- Earth: $\Omega = 7.29 \times 10^{-5} \text{ rad s}^{-1}$
- Mid-latitudes ($\phi = 45^\circ$): $f \approx 1.03 \times 10^{-4} \text{ s}^{-1}$
- Equator ($\phi = 0^\circ$): $f = 0$, so no Coriolis deflection

Explains why winds circulate around pressure centres instead of blowing directly from high to low.



Coriolis deflection of a polar launch, viewed from above the North Pole. The deflection is an apparent force caused by the planet rotating beneath a straight path. Course figure.

The Rossby number

Does planetary rotation dominate a given atmospheric flow?

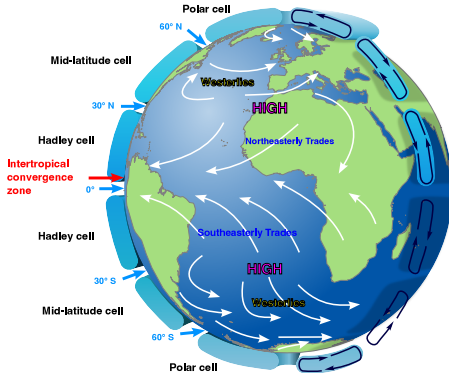
$$Ro = \frac{U}{fL}$$

$Ro \ll 1$: Rotation dominates: large-scale circulation, jet streams (Earth:
 $U \sim 10 \text{ m s}^{-1}$, $L \sim 1000 \text{ km} \rightarrow Ro \sim 0.1$)

$Ro \gg 1$: Rotation unimportant: small-scale convective plumes, tornadoes

The Rossby number sets the dynamic regime and the number of circulation cells.

Circulation cells and rotation rate

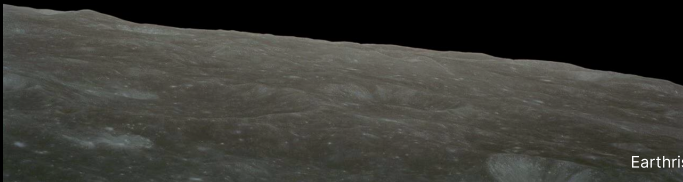


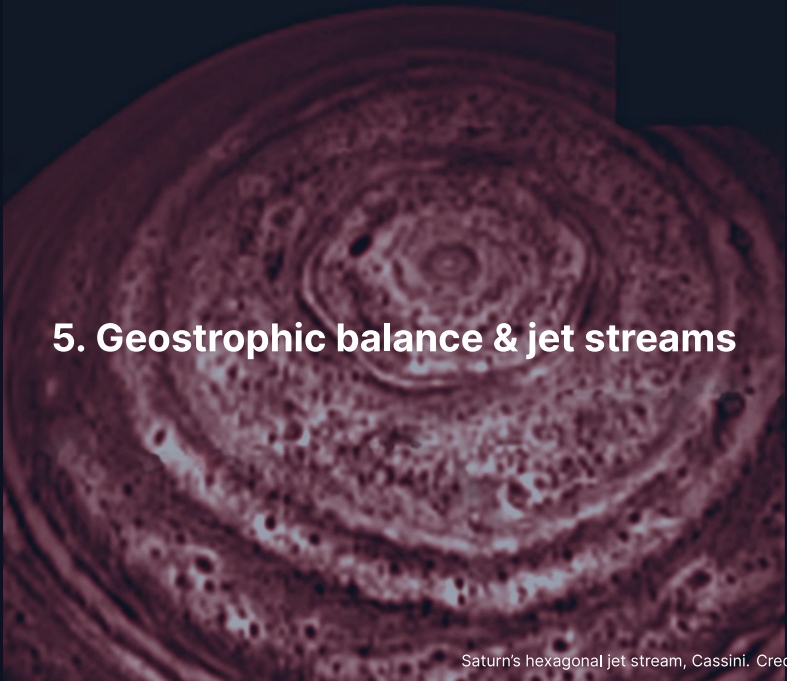
Planetary rotation rate governs the number of circulation cells:

- **Slow rotators** (Venus, Titan): single equator-to-pole Hadley cell per hemisphere ($Ro \gg 1$)
- **Moderate rotators** (Earth): three cells (Hadley, Ferrel, polar)
- **Fast rotators** (Jupiter, Saturn): multiple cells produce banded jets

Credit: Wikimedia Commons, public domain

Break



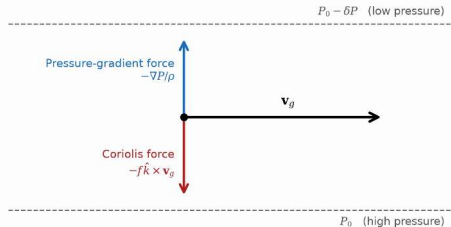
A false-color image of Saturn's atmosphere, showing a prominent hexagonal jet stream. The hexagon is a large, six-sided cloud pattern with a central vortex, surrounded by concentric bands of alternating light and dark colors. The text "5. Geostrophic balance & jet streams" is overlaid in white on the left side of the image.

5. Geostrophic balance & jet streams

Saturn's hexagonal jet stream, Cassini. Credit: NASA/JPL-Caltech/SSI

Geostrophic balance

Geostrophic balance (Northern Hemisphere)



At large scales ($Ro \ll 1$), Coriolis force balances pressure gradient:

$$f \hat{k} \times \mathbf{v}_g = -\frac{1}{\rho} \nabla P$$

$$u_g = -\frac{1}{f\rho} \frac{\partial P}{\partial y}, \quad v_g = \frac{1}{f\rho} \frac{\partial P}{\partial x}$$

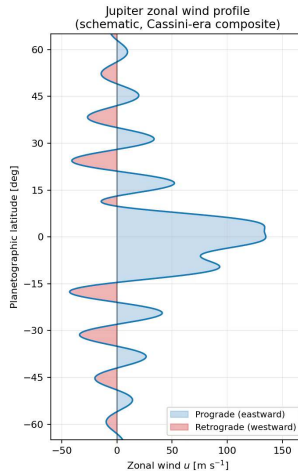
Geostrophic winds blow **parallel to isobars**, rotating cyclonically around lows and anticyclonically around highs.

Jet streams and the thermal wind

Horizontal temperature gradients drive vertical wind shear through the thermal wind relation:

$$\frac{\partial \mathbf{v}_g}{\partial z} \propto \hat{k} \times \nabla T$$

- Steep equator-to-pole ∇T drives fast winds aloft ($\sim 30\text{--}70 \text{ m s}^{-1}$ on Earth)
- Earth hosts subtropical ($\sim 30^\circ$) and polar ($\sim 60^\circ$) jet streams
- Upper-tropospheric jets steer mid-latitude weather systems across the globe



Jupiter's cloud-top zonal winds, schematic after García Melendo & Sánchez-Lavega (2001).
Alternating prograde and retrograde jets reach peak speeds of $\sim 180 \text{ m s}^{-1}$ and extend thousands of kilometres deep. Course figure.

Giant planet banding

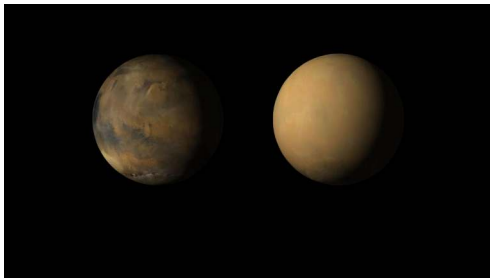
Planetary rotation organises giant-planet atmospheres into alternating bright zones and dark belts:

- **Zones (light):** anticyclonic upwelling, cool ammonia ice cloud tops
- **Belts (dark):** cyclonic downwelling exposing warmer lower layers
- Zonal jets flank band boundaries (Jupiter $\sim 180 \text{ m s}^{-1}$, Saturn $\sim 400 \text{ m s}^{-1}$); decay depth $\sim 1800 \text{ km}$

A close-up, curved view of Jupiter's atmosphere, showing the Great Red Spot as a large, reddish-orange oval storm system. The surrounding atmosphere is a mix of light and dark brownish-grey bands, indicating different cloud layers and chemical compositions. The curvature of the planet is visible at the top and bottom edges of the frame.

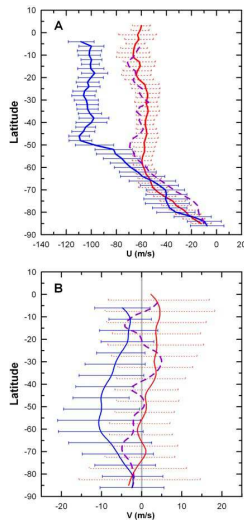
6. Weather & storms in the solar system

Mars: dust storms



Credit: NASA/JPL-Caltech/MSSS

- Dust lofted to 10–30 km can grow into global dust storms
- **Positive feedback loop:** dust absorbs sunlight → heats air → strengthens winds → lofts more dust
- Seasonal CO₂ condensation freezes 25–30% of the atmosphere onto polar caps, swinging the surface pressure between ~5 and 7 mbar



Venus cloud-level winds from VIRTIS on Venus Express: zonal (A) and meridional (B) speed against latitude. Cloud-top zonal winds reach $\sim 105 \text{ m s}^{-1}$, demonstrating that the entire cloud deck super-rotates 60 times faster than the surface. Sánchez-Lavega et al. (2008).

Venus super-rotation: thermal tides

Cloud-top winds circle Venus in ~4 days ($\sim 100 \text{ m s}^{-1}$), 60 times faster than the surface.

- **Puzzle:** sustaining super-rotation requires transporting momentum upward and equatorward
- **Thermal tides:** solar heating generates waves that pump momentum upward
- **Planetary waves:** large-scale waves carry momentum toward the equator

Source: Sánchez-Lavega et al. (2017)

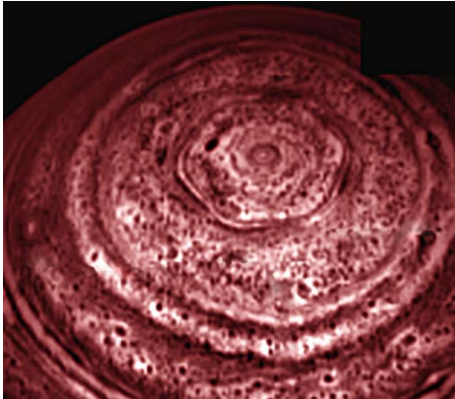
Jupiter: the Great Red Spot



Credit: NASA/JPL-Caltech/SwRI/MSSS (Juno)

- Anticyclonic vortex larger than Earth; continuous records since ~1830
- Peripheral winds reach $\sim 120 \text{ m s}^{-1}$ between opposing zonal jets
- Sustained by merging with smaller vortices and deep H_2O latent heat release
- Slowly shrinking in longitudinal extent over the past century

Saturn: the hexagonal jet stream



Credit: NASA/JPL-Caltech/SSI (Cassini, 2013)

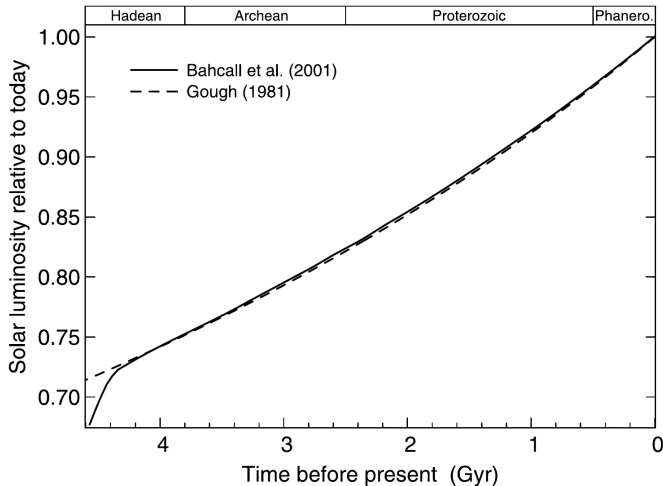
- Persistent hexagonal pattern at $\sim 78^\circ\text{N}$, diameter $\sim 30,000$ km
- Imaged by Voyager (1980–81) and Cassini
- Stationary Rossby wave with azimuthal wavenumber $m = 6$ resonant with the zonal jet
- Saturn also erupts episodic Great White Storms every ~ 30 years (last in 2010)



Neptune's Great Dark Spot, Voyager 2 (1989). Neptune radiates $\sim 2.6\times$ more energy than it receives from the Sun, with internal heat driving $\sim 400 \text{ m s}^{-1}$ winds despite negligible solar heating. Credit: NASA/JPL-Caltech.

An artist's impression of a Snowball Earth, showing a frozen planet with a thin blue ring of liquid water at the equator and a dark blue polar region. The planet is set against a dark blue background with faint stars.

7. Climate evolution & the faint young Sun

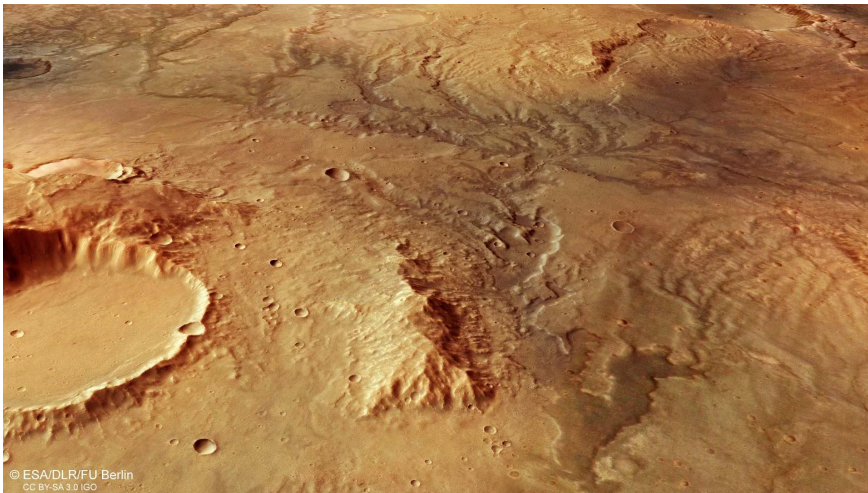


Solar luminosity across geologic time follows $L(t)/L_{\odot} \approx [1 + \frac{2}{5}(1 - t/t_{\odot})]^{-1}$; at $t = 0$, the Sun was ~30% fainter ($0.71 L_{\odot}$) than today. Reproduced from Feulner (2012).

The faint young Sun paradox and solutions

A 30% fainter early Sun implies $T_{\text{eff}} \approx 234$ K, yet geological records require liquid water for >4 Gyr.

- **Geological evidence:** zircons (4.4 Ga), pillow basalts (3.8 Ga), and stromatolites (3.5 Ga) prove ancient oceans
- **Paradox:** with today's atmosphere, early Earth would have been globally frozen
- **Solutions:** stronger greenhouse from elevated CO_2 and biogenic CH_4 , plus lower continental albedo

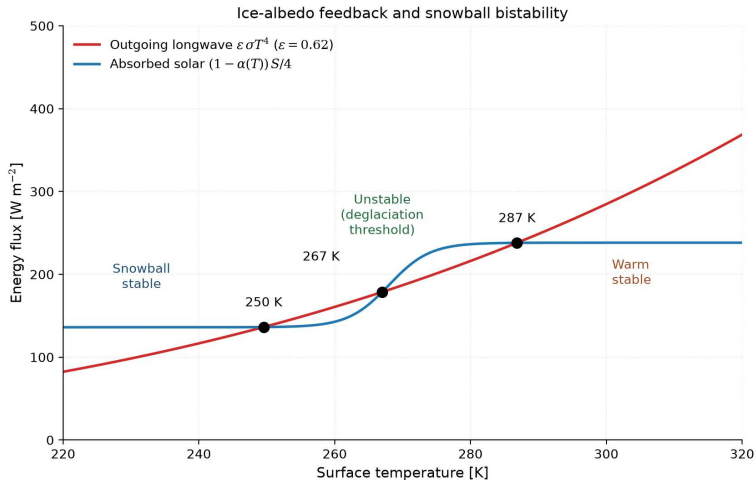


Noachian valley networks, Mars Express HRSC. Fluvial networks require warm, wet conditions on early Mars (>3.7 Ga), an open paradox under the faint young Sun since pure CO_2 condenses into high-albedo clouds. Credit: ESA/DLR/FU Berlin, CC BY-SA 3.0 IGO.

Climate feedbacks and the ice-albedo instability

Atmospheric feedbacks determine whether a planetary climate remains stable or runs away.

- **Water vapour feedback:** warming increases vapour, doubling CO₂ warming (runaway on Venus)
- **Cloud feedback:** competing solar reflection (cooling) and infrared trapping (warming)
- **Ice-albedo feedback:** cooling expands reflective ice; past ~30° latitude, runaway cooling triggers a snowball state



Energy-balance bistability of the ice-albedo feedback. Stable crossings at 250 K and 287 K bracket an unstable threshold at 267 K, separating warm and snowball branches. Course figure.

Snowball Earth

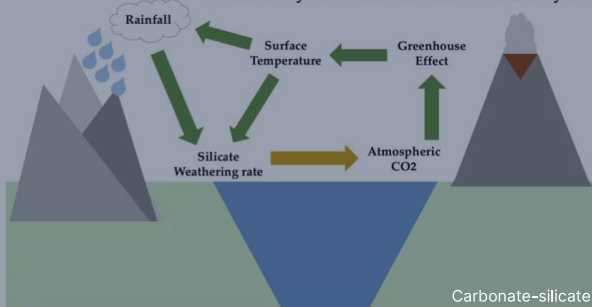
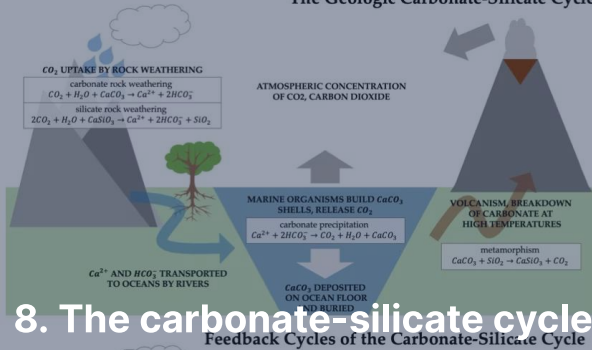


Credit: Oleg Kuznetsov, CC BY-SA 4.0

- Two global glaciations in the Neoproterozoic (~717 and 635 Ma) with tropical glacial deposits
- Surface temperatures dropped below -50°C
- Global ice cover shut down silicate weathering sinks
- Volcanic CO_2 accumulated unchecked over ~10–100 Myr until greenhouse warming melted the ice

Source: Hoffman et al. (2017)

The Geologic Carbonate-Silicate Cycle



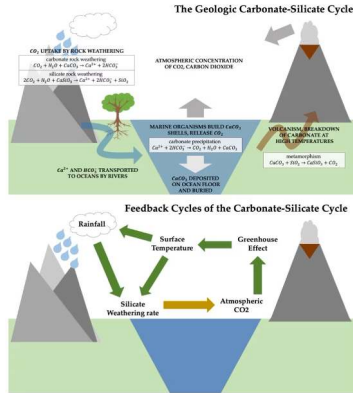
The Urey reaction

Silicate weathering draws atmospheric CO₂ dissolved in rainwater into mineral carbonates:



- Atmospheric CO₂ dissolves in rain forming carbonic acid (H₂CO₃)
- Acid weathers surface silicates, washing Ca²⁺ and HCO₃⁻ to the oceans
- Marine organisms and abiotic processes precipitate carbonate sediments (CaCO₃)
- Net effect: moves carbon from the atmosphere-ocean system into the crust, until subduction and volcanism return it

Volcanic outgassing: the carbon source



Credit: Wikimedia, CC BY-SA 4.0

- Plate tectonics subducts carbonate-rich seafloor into the mantle
- High T and P decompose carbonates, releasing metamorphic CO_2
- **Volcanic outgassing** returns CO_2 to the atmosphere
- Long-term steady state balances weathering sinks against volcanic sources

The negative feedback thermostat

Silicate weathering provides temperature-dependent negative feedback:

- **Warming:** higher rainfall and reaction rates accelerate weathering, drawing down CO_2 and cooling the planet
- **Cooling:** reduced weathering allows volcanic CO_2 to accumulate, strengthening the greenhouse

Operates on ~0.5 Myr timescales, maintaining liquid water for >4 Gyr.

Quantifying the weathering feedback

Silicate weathering rates increase exponentially with temperature to stabilise climate:

$$W(T) = W_0 \exp\left(\frac{T - T_0}{T_e}\right)$$

- With $T_e \approx 10$ K, doubling weathering requires ~ 7 K warming
- Climate perturbations damp on a ~ 0.5 Myr timescale
- Feedback requires liquid water and active volcanism to operate

Source: Walker et al. (1981); Pierrehumbert (2010)

Venus & Mars: failed thermostats

The thermostat requires liquid water and volcanism:

Venus (lost sink):

- Runaway greenhouse boiled oceans
- No water → no weathering sink
- Volcanic CO₂ reached 92 bar

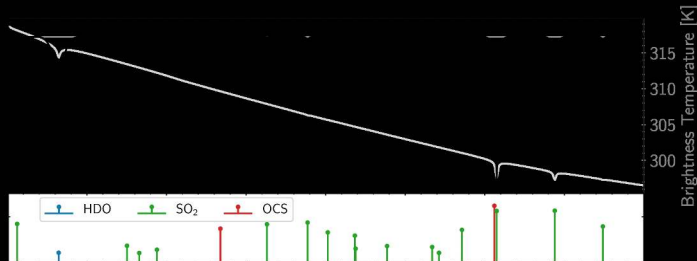
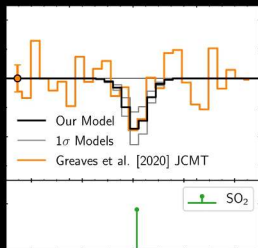
Mars (lost source):

- Volcanism ceased ~3 Ga
- No volcanic return path
- Weathering drawdown and escape left ~6 mbar

Habitability requires active geological cycling, not sunlight alone.

The image features two spheres side-by-side against a dark blue background. The sphere on the left is a realistic representation of Mars, showing its reddish-brown surface with darker patches and polar ice caps. The sphere on the right is a uniform, hazy, light-tan color, representing the planet completely obscured by a global dust storm. Centered over the space between these two spheres is the text '9. Recent advances & outlook' in a bold, white, sans-serif font.

9. Recent advances & outlook



The reported 267 GHz Venus phosphine feature is fit by mesospheric SO₂, a controversy that NASA's DAVINCI descent probe will test in situ. Lincowski et al. (2021).

Summary

1. Clausius-Clapeyron equation predicts saturation vapour pressure and cloud bases
2. Planetary rotation sets circulation cells, from single Hadley loops to banded jets
3. Geostrophic balance and thermal wind govern large-scale winds and storms
4. Faint young Sun paradox is resolved by greenhouse warming and lower albedo
5. Carbonate-silicate weathering provides a ~ 0.5 Myr climate thermostat

Next: Lecture 7

Questions?

Next: Lecture 7, Planetary surfaces: geology, geomorphology & geophysics

Mon 28 Sep, 15:00–17:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

<https://ips.formingworlds.space>